

Course Feedback

Feedback_Jingjing Yang

This was my first FPGA course and it required a significant amount of time, but it was very valuable. Building and testing the system step by step helped avoid rushing everything at the end.

One of the most important things I learned was debugging. For example, always checking whether wires are actually connected in the diagram and making sure signal names match exactly when adding stubs. Small differences in names can easily cause hidden problems. Using tools like ModelSim waveforms to trace signals was also very helpful. I also learned that passing a testbench does not guarantee the design will work correctly on the FPGA board.

The design rules required in the course also encouraged good diagram and design habits. However, due to time pressure it was sometimes difficult to follow every detail perfectly. I imagine in a real team environment there would be even stricter standards.

Overall, the course gave me practical experience with FPGA design and debugging that will be useful in future projects.

Feedback_Joona-Jussi

I think this exercise project was useful because it helped me practice the skills we learned during the course. The tasks took a lot of time to complete, and most of the time I did not understand what to do or how to do it. It was quite difficult for me, but I learned many new things from it.

Improved game features

1. Alien health states and visual feedback

The alien now has multiple health states instead of being defeated with a single hit.

At the start of the game, the alien is green, representing full health. After the first hit it turns yellow, and after the second hit it turns red.

After the third hit, the alien disappears and the player wins the game. The color change provides clear visual feedback about the alien's remaining health.

2. Offensive alien

The alien is made offensive by descending one row each time it reaches the horizontal edge of the screen.

This creates constant pressure for the player because the alien gradually moves closer to the gun. As a result, each round becomes very fast and intense, making the gameplay more challenging.

3. Lose condition and lose effect

A lose condition is added when the alien descends too low. If the alien reaches the same row as the gun tip, the player loses the game.

When this happens, a lose symbol is displayed on the LED matrix: two diagonal parallel red lines, indicating the failure state.

4. Dynamic victory effect

The original victory effect is a static big red heart (40 pixels).

In our improved version, the victory animation becomes dynamic: a line moves left and right while its color cycles through the seven rainbow colors.

This makes the victory screen more vivid and visually appealing compared to the static effect.